Chapter 7. Nonlinear Regression

Lecture 6-4 (polynomials, step functions, basis functions, regression splines, smoothing splines, local regression, GAM)

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1 Basis-Function Regression

1.1 Polynomial regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i.$$

Fitted by OLS on $X_1 = X$, $X_2 = X^2$, …; the target is the fitted function

$$\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \hat{\beta}_2 x_0^2 + \hat{\beta}_3 x_0^3 + \hat{\beta}_4 x_0^4,$$

with pointwise interval $\hat{f}(x_0) \pm t_{1-\alpha/2,\mathrm{df}}\,\mathrm{SE}[\hat{f}(x_0)]$, where $\mathrm{SE}[\hat{f}(x_0)]^2 = \ell_0^\top \widehat{\mathrm{Var}}(\hat{\beta})\ell_0$ with $\ell_0 = (1, x_0, \ldots, x_0^4)^\top$. d is rarely above 3 or 4, or chosen by CV. Auto data: $d = 1$ underfits, $d = 2$ fits well, $d = 14$ is wild.

Limitation. Polynomials are global: distant data affect the local fit and the tails become erratic for large d.

1.2 Step functions

Cut the range of X at $c_1 < \cdots < c_K$:

$$C_0(X) = \mathbb{I}(X < c_1),\ C_1(X) = \mathbb{I}(c_1 \le X < c_2),\ \ldots,\ C_K(X) = \mathbb{I}(c_K \le X),$$

so $C_0 + \cdots + C_K = 1$, and fit $y_i = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$ (C_0 is dropped because of the intercept). β_0 is the mean for $X < c_1$, β_k the difference of bin k. Wage on age with bins < 35, 35–50, 50–65, ≥ 65: the first bin misses the increasing trend. Results depend on the breakpoints.

1.3 Basis functions

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

with fixed b_k: $b_j(x) = x^j$, or $b_j(x) = \mathbb{I}(c_j \le x < c_{j+1})$; least squares, SEs and F-tests apply. Other bases: Fourier, wavelets, splines.

2 Regression Splines

2.1 Piecewise polynomials

Separate low-degree polynomials between **knots**; with one knot c,

$$y_i = \begin{cases} \beta_{01} + \beta_{11} x_i + \beta_{21} x_i^2 + \beta_{31} x_i^3 + \varepsilon_i, & x_i < c, \\ \beta_{02} + \beta_{12} x_i + \beta_{22} x_i^2 + \beta_{32} x_i^3 + \varepsilon_i, & x_i \ge c. \end{cases}$$

Unconstrained pieces jump at the knot; splines impose the maximum continuity.

2.2 Linear splines

Definition: linear spline

A linear spline with knots $\xi_1 < \cdots < \xi_K$ is piecewise linear and continuous at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+1} b_{K+1}(x_i) + \varepsilon_i, \qquad b_1(x) = x, \quad b_{k+1}(x) = (x - \xi_k)_+,$$

with $(x - \xi)_+ = x - \xi$ if $x > \xi$ and 0 otherwise.

$(x - \xi_k)_+$ adds a slope change β_{k+1} at ξ_k.

(a) basis functions (b) linear spline



Figure 1: (a) Truncated power basis, knots ξ_1, ξ_2. (b) A linear-spline fit.

2.3 Cubic splines

Definition: cubic spline and truncated power basis

A cubic spline with knots $\xi_1, \ldots, \xi_K$ is piecewise cubic with continuous derivatives up to order 2 at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i,$$

$$b_1(x) = x,\ b_2(x) = x^2,\ b_3(x) = x^3,\ b_{k+3}(x) = (x - \xi_k)^3_+,\ k = 1, \ldots, K.$$

Theorem: degrees of freedom of a cubic spline

The space of cubic splines with K knots has dimension $K + 4$, and the truncated power basis (with the constant) is a basis.

Proof [Supplement]. $K + 1$ unconstrained cubic pieces have $4(K + 1)$ parameters; continuity of the function, first and second derivatives at each of the K knots imposes $3K$ linear constraints, leaving $4(K + 1) - 3K = K + 4$. Each $(x - \xi_k)^3_+$ is twice continuously differentiable (its first two derivatives vanish at ξ_k) and its third derivative jumps by 6 at ξ_k, so the $K+4$ functions are linearly independent splines. ■

2.4 Natural cubic splines

Splines have high variance at the boundary. A **natural cubic spline** is also *linear* beyond the boundary knots: 4 more constraints (second and third derivatives zero at both ends), so K knots give K degrees of freedom (including the intercept) and narrower boundary bands.

2.5 Number and locations of knots

Place knots at quantiles of X; choose K by CV. Wage data: a degree-15 polynomial is wild at the boundaries, a 15-df natural cubic spline is well behaved.

3 Smoothing Splines

Motivation. Make every data point a knot and control wiggliness by a roughness penalty.

Definition: smoothing spline

Among all twice-differentiable functions g,

$$\min_g \underbrace{\sum_{i=1}^{n}(y_i - g(x_i))^2}_{\text{RSS}} + \lambda \underbrace{\int g''(t)^2\,dt}_{\text{roughness penalty}} .$$

$\lambda = 0$: interpolation; $\lambda \to \infty$: a linear g (zero penalty). λ by CV.

Theorem: the minimizer is a natural cubic spline

The minimizer $\hat{g}_\lambda$ is a natural cubic spline with knots at the unique values of $x_1, \ldots, x_n$ — a shrunken version of the interpolating natural spline. Consequently $\hat{g}_\lambda = (\hat{g}(x_1), \ldots, \hat{g}(x_n))^\top = S_\lambda y$ for an $n \times n$ smoother matrix S_λ.

Idea [Supplement]. The natural cubic spline $\tilde{g}$ interpolating $g(x_i)$ has the same RSS and $\int \tilde{g}''^2 \le \int g''^2$, with equality only if $g = \tilde{g}$ (Exercise 7.3). These splines form an n-dimensional space $g(x) = \sum_j N_j(x)\theta_j$, so the problem becomes $\min_\theta \|y - N\theta\|^2 + \lambda\theta^\top \Omega_N \theta$ with $(\Omega_N)_{jk} = \int N_j'' N_k''$, solved by $\hat{\theta} = (N^\top N + \lambda\Omega_N)^{-1} N^\top y$, a generalized ridge regression. Hence $S_\lambda = N(N^\top N + \lambda\Omega_N)^{-1}N^\top$. ■

Effective degrees of freedom $\mathrm{df}_\lambda = \mathrm{tr}(S_\lambda)$ decrease from n ($\lambda = 0$) to 2 ($\lambda \to \infty$). LOOCV costs one fit: $\mathrm{CV}(\lambda) = \frac{1}{n}\sum_i \{(y_i - \hat{g}_\lambda(x_i))/(1 - \{S_\lambda\}_{ii})\}^2$ (Ch. 3). Wage data: about 6.8 df (by LOOCV) is nearly indistinguishable from 16 df.

4 Local Regression

Weighted linear fits in a sliding window.

Algorithm: local regression at a target point x_0

1. Gather the fraction $s = k/n$ of training points closest to x_0 (the **span**).
2. Assign weights $K_{i0} = K(x_i, x_0)$: zero outside the neighborhood, largest for the closest points (e.g. tricube weights).
3. Fit a weighted least-squares regression $\min_{\beta_0,\beta_1} \sum_i K_{i0}(y_i - \beta_0 - \beta_1 x_i)^2$.
4. The fitted value is $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Choices: weights, local degree (constant, linear, quadratic), and span s (small: wiggly; large: global; by CV). Local linear fits remove the boundary bias of local constant averages. Curse of dimensionality beyond three or four predictors.

5 Generalized Additive Models

Nonlinear in each variable, additive across variables:

$$y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \varepsilon_i.$$

With natural splines a GAM is one least-squares fit; terms may also be smoothing splines, local regression, linear or step functions.

Algorithm: backfitting (for smoothing-spline or local-regression terms)

Initialize $\hat{\beta}_0 = \bar{y}$, $\hat{f}_j = 0$. Repeat until convergence: for $j = 1, \ldots, p$, smooth the partial residuals $r_i = y_i - \hat{\beta}_0 - \sum_{k \neq j} \hat{f}_k(x_{ik})$ against x_{ij} to update $\hat{f}_j$, then center $\hat{f}_j$ to have mean zero.

Example: Wage data

$\texttt{wage} = \beta_0 + f_1(\texttt{year}) + f_2(\texttt{age}) + f_3(\texttt{education}) + \varepsilon$, natural splines for year and age, a constant per education level. With the others fixed, wage rises slightly with year, rises then falls with age, and increases with education.

Pros. Nonlinear f_j without manual transformations; each effect can still be examined with the others fixed; df summarize each f_j. **Cons.** Additivity misses interactions unless added, e.g. $f_{jk}(X_j, X_k)$.

Summary

- Basis expansions (polynomials, steps, splines) stay within least squares; cubic splines have $K + 4$ df, natural cubic splines K.
- Smoothing splines are natural cubic splines with knots at the data; $\mathrm{df}_\lambda = \mathrm{tr} S_\lambda$.
- Local regression: weighted LS in a window; GAMs: additive nonlinear terms (backfitting).

6 Selected exercises

Motivation. ISLR Ex. 7.1 and 7.2, and the least-curvature property behind smoothing splines.

Exercise 7.1 (The truncated power basis gives a cubic spline; ISLR Ex. 7.1)

Let $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$.
(a) Find cubics $f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$ and $f_2(x) = a_2 + b_2 x + c_2 x^2 + d_2 x^3$ with $f = f_1$ on $x \leq \xi$ and $f = f_2$ on $x > \xi$.
(b) Show that $f_1(\xi) = f_2(\xi)$, $f_1'(\xi) = f_2'(\xi)$ and $f_1''(\xi) = f_2''(\xi)$, so f is a cubic spline with knot ξ.

Solution.

(a) For $x \leq \xi$ the truncated term vanishes: $a_1 = \beta_0$, $b_1 = \beta_1$, $c_1 = \beta_2$, $d_1 = \beta_3$. For $x > \xi$, $(x - \xi)^3 = x^3 - 3\xi x^2 + 3\xi^2 x - \xi^3$, so

$$a_2 = \beta_0 - \beta_4 \xi^3, \quad b_2 = \beta_1 + 3\beta_4 \xi^2, \quad c_2 = \beta_2 - 3\beta_4 \xi, \quad d_2 = \beta_3 + \beta_4.$$

(b) As polynomials, $f_2 - f_1 = \beta_4 (x - \xi)^3$, with derivatives $3\beta_4 (x - \xi)^2$ and $6\beta_4 (x - \xi)$. All three vanish at $x = \xi$. (The third derivative jumps by $6\beta_4$.) ■

Exercise 7.2 (Which penalty, which limit; ISLR Ex. 7.2)

Let $x_1 < \cdots < x_n$ be distinct points in $[a, b]$, $n \geq 3$, and

$$\hat{g} = \arg\min_g \sum_{i=1}^{n} (y_i - g(x_i))^2 + \lambda \int_a^b \left(g^{(m)}(x)\right)^2 dx$$

over functions with continuous mth derivative ($g^{(0)} = g$). Interpreting $\lambda = \infty$ as the constraint that the penalty vanishes, describe $\hat{g}$ for $\lambda = \infty$ and $m = 0, 1, 2, 3$, and for $\lambda = 0$, $m = 3$.

Solution.

Step 1. A continuous nonnegative function with zero integral vanishes identically, so $\int_a^b (g^{(m)})^2 = 0$ iff $g^{(m)} \equiv 0$ on $[a, b]$. For $m = 0$ this means $g \equiv 0$; for $m \geq 1$, integrating m times, it means that g is a polynomial of degree at most $m - 1$.

Step 2. With $\lambda = \infty$, $\hat{g}$ minimizes RSS over that class:

(a) $m = 0$: $\hat{g} \equiv 0$.

(b) $m = 1$: constants; $\sum_i (y_i - c)^2$ is minimized at $c = \bar{y}$, so $\hat{g} \equiv \bar{y}$.

(c) $m = 2$: lines; $\hat{g}$ is the least-squares line.

(d) $m = 3$: quadratics; $\hat{g}$ is the least-squares quadratic, unique because three or more distinct x_i make the design $(1, x_i, x_i^2)$ of full column rank.

Step 3. With $\lambda = 0$, $m = 3$, every smooth g with $g(x_i) = y_i$ (e.g. the interpolating polynomial of degree $n - 1$) attains the minimum RSS $= 0$: $\hat{g}$ interpolates and is not unique. ■

Exercise 7.3 (The natural cubic spline has least curvature)

Let $a < x_1 < \cdots < x_n < b$ with $n \geq 2$, and let $\tilde{g}$ be a natural cubic spline with knots $x_1, \ldots, x_n$ (linear on $[a, x_1]$ and $[x_n, b]$). Let g be any twice continuously differentiable function on $[a, b]$ with $g(x_i) = \tilde{g}(x_i)$ for all i. Show that $\int_a^b g''^2 \geq \int_a^b \tilde{g}''^2$, with equality iff $g = \tilde{g}$. Deduce that the smoothing-spline minimizer is a natural cubic spline with knots at the x_i.

Solution [Supplement]. Let $h = g - \tilde{g}$; it is twice continuously differentiable and $h(x_i) = 0$ for all i.

Step 1. $\tilde{g}''$ is continuous and piecewise linear, hence absolutely continuous with derivative $\tilde{g}'''$ almost everywhere, where $\tilde{g}'''$ equals a constant c_j on (x_j, x_{j+1}) and 0 on (a, x_1) and (x_n, b). Integration by parts gives

$$\int_a^b \tilde{g}'' h'' = \left[\tilde{g}'' h'\right]_a^b - \int_a^b \tilde{g}''' h'.$$

The boundary term vanishes because $\tilde{g}$ is linear near a and b, so $\tilde{g}''(a) = \tilde{g}''(b) = 0$.

Step 2. $\displaystyle\int_a^b \tilde{g}''' h' = \sum_{j=1}^{n-1} c_j \int_{x_j}^{x_{j+1}} h' = \sum_{j=1}^{n-1} c_j \{h(x_{j+1}) - h(x_j)\} = 0$. Hence $\int \tilde{g}'' h'' = 0$.

Step 3. Therefore

$$\int_a^b g''^2 = \int_a^b (\tilde{g}'' + h'')^2 = \int_a^b \tilde{g}''^2 + \int_a^b h''^2 \;\geq\; \int_a^b \tilde{g}''^2.$$

Step 4. Equality forces $\int h''^2 = 0$, so $h'' \equiv 0$ (continuity) and h is linear. A linear function vanishing at $n \geq 2$ points is 0, so $g = \tilde{g}$.

Step 5 (consequence). For any candidate g, the natural cubic spline $\tilde{g}$ interpolating $g(x_i)$ has the same RSS and, by Steps 3–4, a strictly smaller penalty unless $g = \tilde{g}$. ■